The Hamiltonian of the tight binding model is

$$H = -t \sum_{<i,j>,\eta} c_{i,\eta}^{+} c_{j,\eta} - J_H \sum_{i} c_{i,\eta}^{+} \sigma_{\eta\eta'} \cdot S_i c_{i,\eta'} .$$

It is based on a 32×32 skyrmion structure through https://juspin.de/wp-content/spirit/. The parameters are $t = -1$, $\mu = 1$, $\eta = 0.1$, $T = 1.5$ and the coupling constant J_H is not fixed. The results are shown in Fig.1

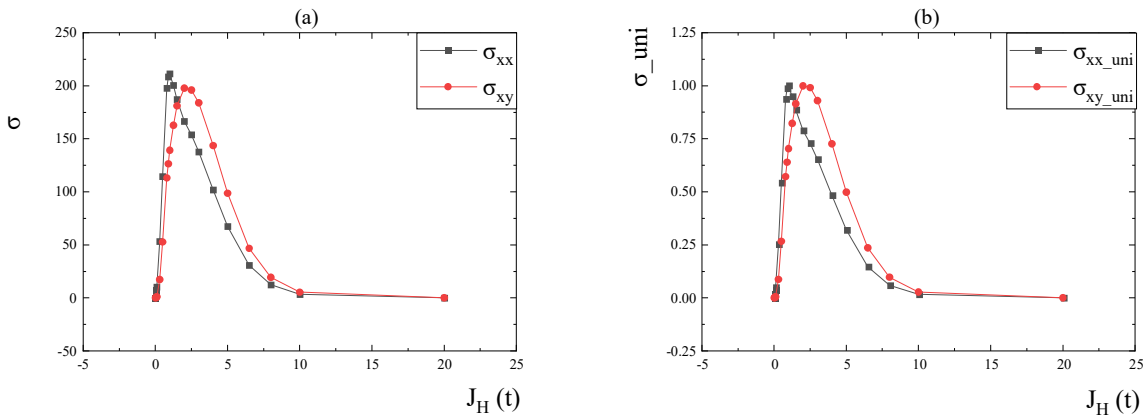

Fig.1 (a)Original values of conductivites vs J_H. (b)Normalized values of conductivites vs J_H.

The parameters $t = -1$, $\eta = 0.1$, $T = 1.5$ and chemical potential is not fixed. The results are shown in Fig.2.

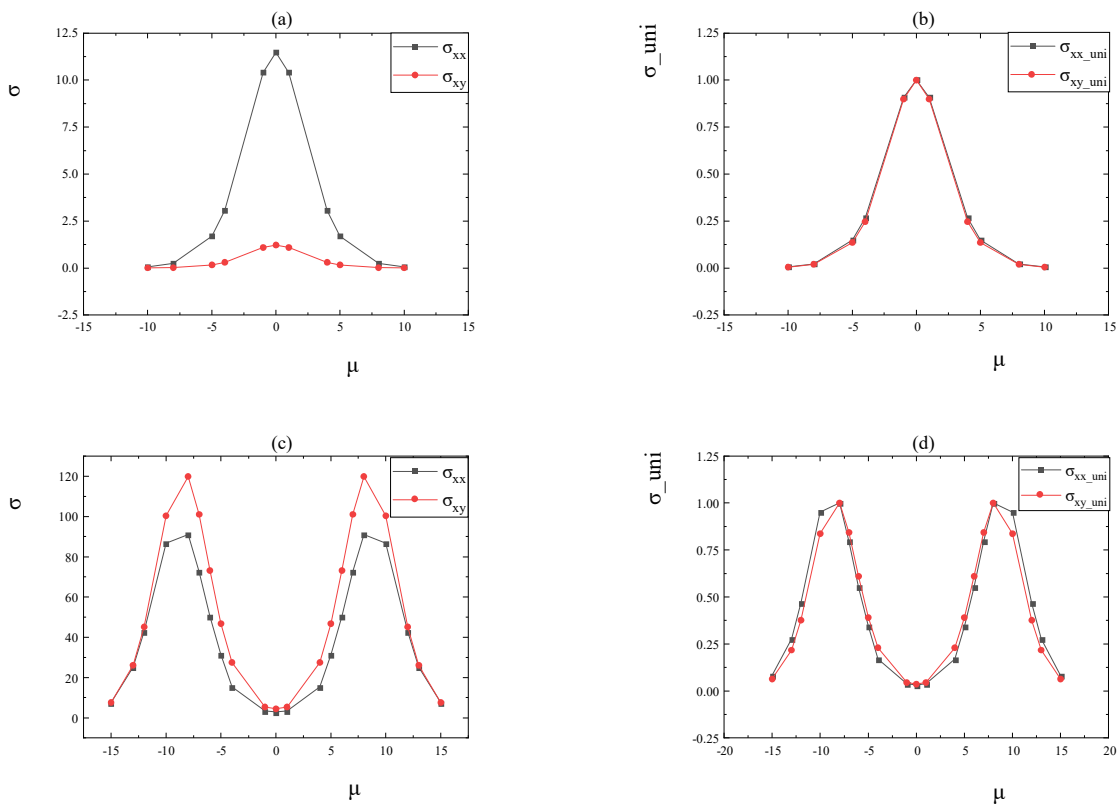

Fig.2 (a)(b) are weak coupling ($J_H = -0.1$) case. (c)(d) are strong coupling ($J_H = -10$) case.